Introduction to Electrical Engineering

Experiment 3 **Spring 2019**

Verification of Kirchhoff's Laws

It is important to remember that Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) are applicable to any electrical circuit, irrespective of the characteristics of the circuit elements including the sources. Just to emphasise this fact, the circuit we will use in this experiment will consist of three linear resistors, a linear capacitor and a diode (nonlinear resistor), with an independent voltage source applied from the Wave Generator of the DSO, as shown in **Fig. 3.1**. **a**, **b**, **c** and **d** denote four points on the circuit, and the Ground symbol (⏚) indicates the common reference node for voltages. The voltages at the four points **a**, **b**, **c**, **d** with respect to Ground will be denoted by v_a, v_b, v_c and v_d, and the voltages between any two points by a variable with two subscripts; e.g. $v_{ab} = v_a - v_b$, $v_{ac} = v_a - v_c$, $v_{bc} = v_b - v_c$ and so on.

Verification of KCL: This will be done for node **c**, and will thus involve the measurement of the instantaneous values of the currents i_1, i_2 and i_3, as indicated in **Fig. 3.1**, at two instants of time $t_1 = -0.4$ms and $t_2 = 0.3$ms, as indicated in **Fig. 3.2**, followed by verifying that the equation $i_1 - i_2 - i_3 = 0$ is satisfied both at t_1 and at t_2. The values of t_1 and t_2 have been so chosen that they give reasonable values of all the voltages of the circuit.

As all our measurements have to be done with a Digital Storage Oscilloscope (DSO), which is basically a voltage-measuring instrument, the values of the currents i_1, i_2 and i_3 will have to be obtained by measuring the voltages v_{ab}, v_c and v_d, and then invoking Ohm's Law which gives $i_1 = v_{ab} / R_1$, $i_2 = v_c / R_2$ and $i_3 = v_d / R_3$. If, for the sake of simplicity, we choose $R_1 = R_2 = R_3$, then the equation $i_1 - i_2 - i_3 = 0$ reduces to $v_{ab} - v_c - v_d = 0$.

Verification of KVL: This will be done for the loop **Gnd**-**a**-**b**-**c**-**Gnd**, and will thus involve the measurement of the instantaneous values of the voltages v_a, v_{ab}, v_{bc} and v_c at the chosen instants of time t_1 and t_2, followed by the verification that they satisfy the equation $v_a - v_{ab} - v_{bc} - v_c = 0$ both at t_1 and at t_2.

The waveform of the voltage v_a applied to the circuit will be a 1kHz **triangular** waveform with its lower tip at the level 0V and its upper tip at the level 4V, as shown in **Fig. 3.2**.

Experimental Steps

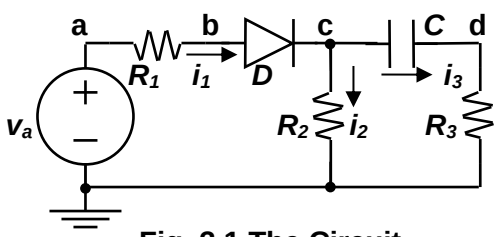

Fig. 3.1 The Circuit

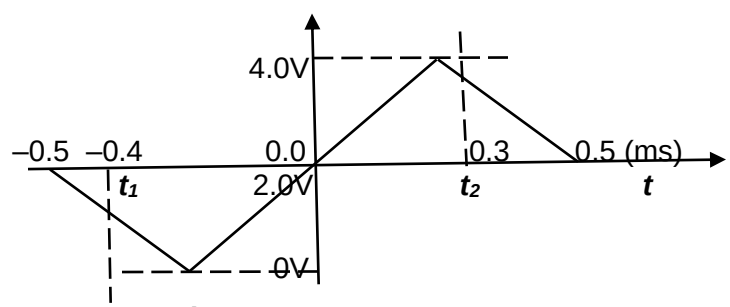

Fig. 3.2 Wavegen Output v_a

1. Assemble the circuit given in **Fig. 3.1** on the breadboard using three ¼W 1.20kΩ resistors as R_1, R_2 and R_3, one 0.1μF polyester capacitor as C, and one 1N4148 diode as D. Connect both the **Wavegen** output and the Channel 1 input of the **DSO** to point ***a*** of the circuit, and adjust the voltage v_a to have a **triangular** waveform with frequency 1kHz and **peak-to-peak** output voltage 4.0V with an **offset** +2.0V, so that the waveform of v_a is as shown in **Fig. 3.2**. Do not rely on the **Wavegen** settings; adjust them suitably so that the lower and upper tips of the waveform are exactly 0 and 4V, as measured by cursors.

2. Switch on the DSO and set the Trigger **Source** of the DSO as **Wavegen** with rising slope, so that the displayed waveforms will all be synchronised with the applied voltage v_a. The advantage of this choice of the trigger source is that triggering does not get disturbed when we connect different voltages to the DSO channels for making measurements. Adjust the DSO controls so that the display on Channel 1 shows **one full cycle** of the v_a waveform, with the time origin at the **centre** of the screen, as shown in **Fig. 3.2**.

3. Turn on the **cursors** X1 and Y1 in the Manual mode. Select them one by one and align them with the time and voltage axes using the cursor control knob, indicated by the highlighted value of the cursor positions becoming **exactly zero** for the X1 cursor and **exactly** 2.0V for the Y1 cursor, which now form the coordinate axes for all the graphs as in **Fig. 3.2**. The remaining cursors X2 and Y2 will be used to mark the time instant of measurement and the measured value of the waveform respectively, as described below.

Measurement using Cursors in Manual Mode

4. Select the Manual Mode for cursors. Select cursor X2 and move it to mark $t = t_1$ using the cursor control knob, indicated by the values of X2 and ΔX both becoming exactly equal to t_1.

5. Connect Channel 2 Input of the DSO to the point **b** by a BNC cable, and select both Channels and the Mathematical function CH1 – CH2, so that v_a, v_b and v_{ab} are simultaneously displayed on the screen.

6. Measure the **instantaneous** value of v_a at $t = t_1$ by adjusting the position of cursor Y2 so that it passes through the point of intersection of cursor X2 with the waveform of v_a displayed on Channel 1, and then reading the value of Y2 from the digital readout provided by the DSO.

7. Measure the **instantaneous** value of v_{ab} at $t = t_1$ by displaying the waveforms of v_b and v_{ab} on CH2 and the Math output respectively and adjusting cursor Y2 as done in step 6 for measuring v_a at $t = t_1$.

8. Repeat steps **5**, **6** and **7** with point ***c*** connected to CH1, and using the Mathematical function CH2 – CH1, so as to display v_c, v_b and v_{cb} on the screen, and hence obtain the values of v_c and v_{cb} at $t = t_1$.

9. Repeat steps **5**, **6** and **7** with point ***d*** connected to Channel 2, to obtain the value of v_d at $t = t_1$.

10. Tabulate the measured values and the calculated values of $v_{ab} - v_c - v_d$ and $v_a - v_{ab} - v_{bc} - v_c$.

Measurement using Cursors in Track Waveform Mode

11. In the method outlined above, the accuracy of the final measurement is limited by eye-estimation error involved in the positioning of the Y2 cursor so that it passes through the point of intersection of cursor X2 with the waveform under measurement. The Track Waveform Mode of Cursors reduces this error by making each Y cursor to automatically track the movement of the corresponding X cursor so that the X and Y cursors always intersect on the chosen Source Waveform. Select this mode for cursors, with both X1-Y1 and X2-Y2 tracking the Channel 1 signal (v_a). As X1 and Y1 have already been aligned with the voltage and time axes, there will be no change in their positions. And as X2 was set to mark $t = t_1$, Y2 will now give the value of the chosen Waveform v_a at $t = t_1$.

12. Connect Channel 2 Input of the DSO to the point **b** by a BNC cable. Choose the Math output (v_{ab}) as the Source Waveform for the tracking of X2-Y2 and read the displayed value of Y2 to obtain the **instantaneous** value of v_{ab} at $t = t_1$.

13. Repeat step **12** with Channel 1 Input connected to the point ***c***, selecting the Channel 1 signal (v_c) and the Math output (v_{cb}) as the Source Waveform for the tracking of X2-Y2 one by one and read the displayed value of Y2 to obtain the **instantaneous** values of v_c and v_{cb} at $t = t_1$.

14. Repeat step **12** with Channel 2 Input connected to the point ***d***, selecting the Channel 2 signal (v_d) as the Source Waveform for the tracking of X2-Y2 and read the displayed value of Y2 to obtain the **instantaneous** value of v_d at $t = t_1$.

15. Tabulate the measured values and the calculated values of $v_{ab} - v_c - v_d$ and $v_a - v_{ab} - v_{bc} - v_c$ at $t = t_1$.

16. Perform the measurements at $t = t_2$, by repeating steps **11-15** with cursor X2 set at $t = t_2$.

17. Compare the magnitude of $v_{ab} - v_c - v_d$ with those of smallest and largest of the individual voltages, both at $t = t_1$ and at $t = t_2$ and comment on the errors observed in the two methods of measurement.

18. Compare the magnitude of $v_a - v_{ab} - v_{bc} - v_c$ with that of the individual voltages, both at $t = t_1$ and at $t = t_2$ and comment on the errors observed in the two methods of measurement both at $t = t_1$ and at $t = t_2$.

RESULT TABLE

	v_a	v_{ab}	v_c	v_{cb}	v_d	$v_{ab} - v_c - v_d$	$v_a - v_{ab} - v_{bc} - v_c$
t_1							
t_2							